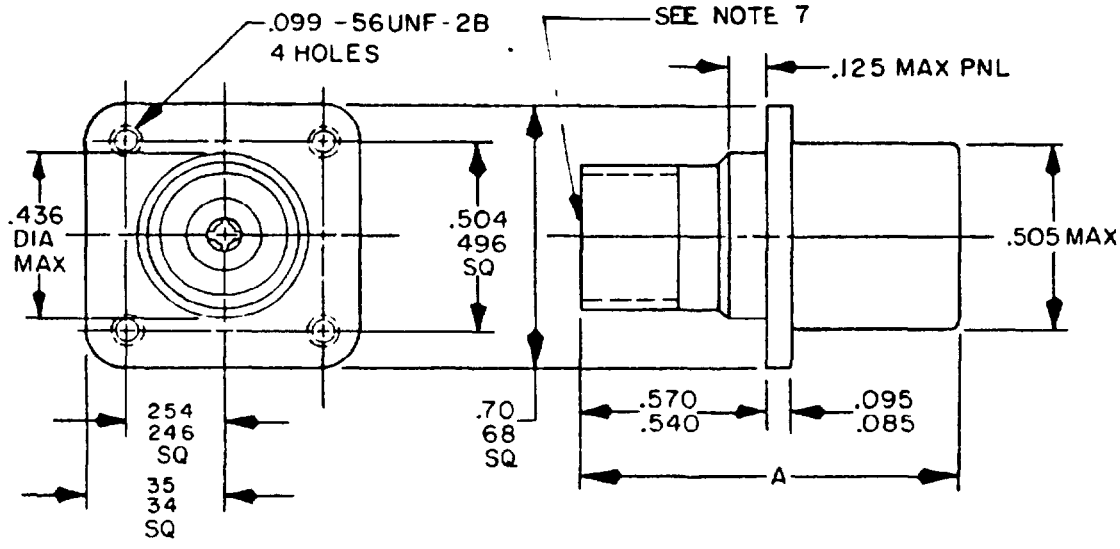
MILITARY SPECIFICATION SHEET

CONNECTORS, RECEPTACLE, ELECTRICAL, COAXIAL, RADIO FREQUENCY (SERIES TNC (CABLED) - SOCKET CONTACT, FLANGE MOUNTED, CLASS 2) FOR SEMI-RIGID CABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-39012.



Inches	mm	Inches	mm
.078	1.98	.436	11.07
.085	2.16	.496	12.60
.095	2.41	.504	12.80
.099	2.59	.505	12.83
.125	3.18	.540	13.72
.246	6.25	.570	14.48
.254	6.45	.68	17.3
.34	8.6	.70	17.8
.35	8.9	1.437	22.35

MARKING IMPLEMENTATION DATE, CATEGORY B, SEE TABLE IV

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Receptacle not recommended for use on panels with greater than .078 (1.98 mm) maximum thickness.
4. Wrench flats are to accommodate standard wrench openings in accordance with FED-STD-428, appendix 10.
5. Dimension A (see table I and III) defines the maximum length of the connector when assembled to the appropriate cable.
6. All undimensioned pictorial configurations are for reference purposes only.
7. Series TNC, socket contact interface per MIL-STD-348

FIGURE 1. General configuration

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

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TABLE I. Dash number and applicable cable.

Dash No	Applicable cable No	Dimension A
CATEGORY A - Field Servicable (no special tools required)		
0001 1/ 2/	M17/130 - RG402	1.203
0002 1/ 3/	M17/129 - RG401	1.250

1/ These connectors have captivated center contacts.
2/ This connector accommodates all cables listed on MIL-C-39012/130.
3/ This connector accommodates all cables listed on MIL-C-39012/129.

ENGINEERING DATA

Nominal impedance: 50 ohms.

Frequency range: 0 to 15,000 MHz.

Voltage rating 500 volts rms, maximum working voltage at sea level. 125 volts rms, maximum at 70,000 feet.

Temperature rating -65°C to +105°C

REQUIRMENTS:

Dimensions and construction: See figure 1.

Force to engage and disengage

Longitudinal force - Not applicable.
Torque - 2 inch-pounds, maximum.

Coupling proof torque Not applicable.

Inspection conditions

Coupling torque - 4 to 6 inch-pounds.

Mating characteristics See MIL-STD-348.

Center contact (socket)

Oversize test pin - .057 diameter minimum (nonclosed entry contacts only).

Insertion depth - .125 minimum.

Number of insertions - 1.

Insertion force test - Steel test pin diameter .054 minimum.

Test pin finish - 16 microinches.
Insertion force - 2 pounds, maximum.

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Withdrawal force test Steel test pin diameter .052 maximum.

Withdrawal force - 2 ounces, minimum
Test pin finish - 16 microinches.

Hermetic seal Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance Method 302, test condition B, MIL-STD-202. 5,000 megohms, minimum.

Center contact retention 6 pounds, minimum axial force. Applicable to captivated-center-contact connectors only.

Corrosion (salt spray) Method 101, test condition B, MIL-STD-202.

Voltage standing wave ratio (VSWR): From 2,000 to 15,000 MHz, or approximately 80 percent of upper cutoff frequency of the cable, whichever is lower, 1.35 maximum.

Swept frequency VSWR test setup

Item 6 - VSWR shall be less than 1.02 + .003 F (F in GHz).

Item 16 - VSWR shall be less than 1.02 + .003 F (F in GHz).

Second step of VSWR checkout procedure - VSWR shall be less than 1.06 + .007 F (F in GHz).

Group B inspection - VSWR shall be less than 1.05 + .005 F (F in GHz).

Qualification and group C inspection - VSWR shall not exceed 1.10.

Connector durability 500 cycles at 12 cycles per minute maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance In milliohms maximum

	Initial	After environment
Center contact	1.5	2.0
Outer contact	0.2	Not applicable
Outer cable conductor to body	0.5	Not applicable

Dielectric withstanding voltage method 301 of MIL-STD-202. 1,500 volts rms, minimum at sea level.

Vibration, high frequency: Method 204, MIL-STD-202, test condition B, except the method of mounting shall be approved by the qualifying activity.

Shock Method 213 of MIL-STD-202, test condition I.

Thermal shock Method 107, test condition B, MIL-STD-202, except test high temperature shall be 115°C.

Moisture resistance Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level

Voltage - 375 volts, minimum.
Altitude - 70,000 feet.

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RF high potential withstanding voltage:

Voltage and frequency 1,000 volts rms at a frequency between 5 and 7.5 MHz, inclusive.

Leakage current. Not applicable.

Cable retention force·

Cable	Pounds (min.)	Torque inch-pounds 1/
M17/130 - RG402	60	3.5
M17/129 - RG401	90	5

1/ Torque shall be applied one cable diameter from the end of the connector. Bend not applicable.

Coupling mechanism retention force: Not applicable.

RF leakage: -60 dB, minimum, tested at a frequency between 2 and 3 GHz.

Insertion loss.

$.06\sqrt{F\ (GHz)}$ dB, maximum tested at 3 GHz.

Part number: M39012/116 (and dash number from table I or "B" number from table III).

TABLE II. Group qualification.

Group	Submission and qualification of any of the following connectors	Qualifies the following connectors
1	M39012/113-0001 M39012/113-0002 M39012/115-0001 M39012/115-0002 M39012/116-0001 M39012/116-0002	M39012/113-0001 M39012/113-0002 M39012/115-0001 M39012/115-0002 M39012/116-0001 M39012/116-0002

TABLE III. Category B-nonfield replaceable (special tools may be required).

NOT FOR AIR FORCE OR NAVY USE. FOR OEM USE ONLY!

M39012/116B 1/ 2/	Applicable Cable No.	Dimension A
0005	M17/130 - RG402	1.203
0006	M17/129 - RG401	1.250

1/ These connectors have captivated center contacts.
2/ For field maintenance purposes these connectors shall be replaced with a category A connector.

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TABLE IV. Cross-reference of part numbers.

Part number 1/ M39012/116B	Superseded part numbers M39012/116-
0005	0005, 0003
0006	0006, 0004

1/ The new "B" part numbers will be required marking 6 months after the date of this specification. The previous part number may be used in the interim.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy -
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AT, ME
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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